

Grounded Agents: Entry Points, Domain Value, and the Transfer Operator

Application Paper II of the *Grounded Agents* series
A Companion to the Theory Paper

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Abstract

We apply the inference-first framework to the entry-point problem: where, how, and why a resource-constrained entrant should close its first epistemic loop. We formalize loop cost, data propriety, commercial value, and the domain value DV; we prove that the cheapest general entry point is a tabletop manipulation loop; and we derive a ranking of domains by DV. We then formalize the transfer coefficient T , its three regimes, its four determinants, and its consequences for the industrial shape of the grounded-agent economy: raising T is the generalization problem, and the platform winner comes from transfer, not scale. Finally we formalize the structural inversion—the shift of the value layer from the model layer to the loop layer—and distinguish futile from non-futile strategy. A companion appendix reports long-range market projections and reconciles them with the framework’s own criteria: rank by market size, filter by loop cost, choose by data propriety, win by transfer. All results are conditional on the axioms of Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*.

Remark 0.1 (Reading). Theorem numbers in this paper are local. The framework, definitions, and proofs of the results recalled in Section 2 live in Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*; the industrial-structure results used below are developed in Application Paper I.

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1 Introduction

Application Paper I answered *where* value accrues in the grounded-agent economy: to the closed loop and its grounded data. This paper answers a different, complementary question: given resource constraints, where should an entrant *start*? The two questions are inverses of each other. The integrator moat says where value accrues; the entry point says where to begin. Both follow from the same principle: grounded data is the proprietary asset, and the cheapest way to acquire it is to close the loop in a bounded environment.

Three questions organize the paper. First: what is a well-chosen entry point, and how do we rank entry points by value (Section 3)? Second: once the first loop is closed, what governs whether it can be extended to adjacent environments (Section 4)? Third: what does the resulting industry look like, and what does that imply for strategy (Section 5, Section 6)?

A methodological note. The paper’s formal results are theorems from the axioms: they are conditional on the framework. The appendix (Section I) introduces empirical content—long-range market projections—which are used to instantiate the components of DV but are not themselves framework consequences. Sections I and 13 therefore separate the framework’s claims from its empirical readings as sharply as the text can.

2 Recalled Framework

This paper applies the framework of Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics* and the industrial structure of Application Paper I. We recall the definitions and results used below; proofs are omitted here.

Definition 2.1 (Agent and Grounding). An *agent* is a tuple $\mathcal{A} = (\mathcal{I}, \mathcal{S}, \mu)$ with inference system $\mathcal{I} = (\mathcal{C}, \vdash, \sigma)$, sensory states $\mathcal{S}$, and grounding map μ . The belief set at s is $\mathcal{B}_{\mathcal{A}}(s) = \sigma(\mu(s))$. The *grounding coefficient* $G(\mathcal{A})$ measures the fraction of belief contents grounded in $\mathcal{A}$ ’s own sensory history.

Definition 2.2 (Environment and Indexed Grounding). An *environment* $\mathcal{E}$ is a structured domain of interaction. The *environment-indexed grounding coefficient* is $G_{\mathcal{E}}(\mathcal{A}) = \inf_{s \in \mathcal{S}_{\mathcal{E}}} G(\mathcal{A}, s)$, over sensory states produced by interaction with $\mathcal{E}$.

Definition 2.3 (Loop Closure and Grounded Data). Agent $\mathcal{A}$ *closes the loop* in $\mathcal{E}$ if it has a self-sourced observation operator with respect to $\mathcal{E}$ and $G_{\mathcal{E}}(\mathcal{A}) > 0$. Its *grounded data* is

$$\mathcal{D}_{\mathcal{A}}(\mathcal{E}) = \bigcup_{s \in \mathcal{S}_{\mathcal{E}}} \mathcal{B}_{\mathcal{A}}(s) \setminus \nu(\mathcal{C}_{\text{public}}),$$

the belief contents in $\mathcal{E}$ not derivable from public corpora.

Theorem 2.4 (Integrator Moat). *An integrator that closes the loop in $\mathcal{E}$ acquires a resource $\mathcal{D}_{\mathcal{A}}(\mathcal{E})$ that is not obtainable by component purchase.*

Theorem 2.5 (Grounding Cannot Be Purchased). *No purchase of components yields $G_{\mathcal{E}}(\mathcal{B}) > 0$ for an agent $\mathcal{B}$ without $\mathcal{B}$ ’s own interaction with $\mathcal{E}$.*

Theorem 2.6 (Vertical Specificity and Generality Delay). *Grounding is environment-specific: $G_{\mathcal{E}_1} > 0$ does not imply $G_{\mathcal{E}_2} > 0$. Consequently the general-purpose tinkering arrival time is bounded below by the maximum of the specialized arrival times, and the robotics era is staggered into domain-specific eras.*

Remark 2.7. Throughout, “the moat,” “staggered eras,” and “environment-specificity” refer to Theorem 2.4, Theorem 2.6, and the vertical-specificity result of Application Paper I. All are proved there or in Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*.

3 Entry Points and Domain Value

The tinkering threshold of Application Paper I answers *when* a closed-loop system becomes affordable. It does not answer *where* to close the first loop. This section formalizes the entry-point problem and derives a ranking of domains by framework-internal criteria.

3.1 Entry Points

Definition 3.1 (Entry Point Loop). An *entry point loop* for an agent $\mathcal{A}$ is a narrow loop $\mathcal{L} \subseteq \mathcal{E}$ in an environment $\mathcal{E}$ such that:

- (E1) $\mathcal{A}$ closes the loop in $\mathcal{E}$ (Definition 2.3),
- (E2) the loop architecture of $\mathcal{A}$ is general—it can be applied to other environments without structural modification,
- (E3) the grounded data $\mathcal{D}_{\mathcal{A}}(\mathcal{E})$ transfers, in whole or in part, to adjacent environments.

Definition 3.2 (Loop Cost). The *loop cost* of an entry point is the total cost of closing it:

$$c_{\text{loop}}(\mathcal{L}) = c_{\text{hw}}(\mathcal{L}) + c_{\text{env}}(\mathcal{L}) + c_{\text{int}}(\mathcal{L}),$$

where c_{hw} is hardware cost, c_{env} is environment cost (lease, setup, safety certification), and c_{int} is integration cost.

Definition 3.3 (Data Propriety). The *data propriety* of an entry point is the fraction of $\mathcal{D}_{\mathcal{A}}(\mathcal{E})$ that cannot be derived from public corpora:

$$P(\mathcal{L}) = \frac{|\mathcal{D}_{\mathcal{A}}(\mathcal{E})|}{|\mathcal{D}_{\mathcal{A}}(\mathcal{E})| + |\nu(\mathcal{C}_{\text{public}})|}.$$

Definition 3.4 (Commercial Value). The *commercial value* of an entry point is

$$V(\mathcal{L}) = w_{\text{labor}} \cdot \Delta_{\text{labor}} + w_{\text{market}} \cdot M_{\text{market}},$$

where Δ_{labor} is the labor cost replaced per unit, and M_{market} is the addressable market size.

Definition 3.5 (Transfer Coefficient). The *transfer coefficient* of an entry point $\mathcal{L}$ in environment $\mathcal{E}$ to an adjacent environment $\mathcal{E}'$ is

$$T(\mathcal{L}, \mathcal{E}') = \frac{G_{\mathcal{E}'}(\mathcal{A} \mid \mathcal{D}_{\mathcal{A}}(\mathcal{E}))}{G_{\mathcal{E}'}(\mathcal{A})},$$

the ratio of grounding in $\mathcal{E}'$ achievable from data learned in $\mathcal{E}$ to grounding in $\mathcal{E}'$ achievable from scratch.

Definition 3.6 (Domain Value). The *domain value* of an entry point is

$$\text{DV}(\mathcal{L}) = \frac{V(\mathcal{L}) \cdot P(\mathcal{L}) \cdot \max_{\mathcal{E}'} T(\mathcal{L}, \mathcal{E}')}{c_{\text{loop}}(\mathcal{L})}.$$

Remark 3.7. The transfer term in DV rewards entry points whose grounded data is a seed rather than a terminal asset: value compounds only where the loop extends. This is the formal content of “win by transfer.”

3.2 The Cheapest Entry Point

Theorem 3.8 (Cheapest General Entry Point). *Among entry points satisfying Definition 3.1, the minimum loop cost is achieved by a tabletop manipulation loop with a low-cost robotic arm.*

Proof. By Definition 3.2, minimizing c_{loop} requires minimizing each of c_{hw} , c_{env} , and c_{int} . Tabletop manipulation minimizes each:

- (a) c_{hw} : a low-cost robotic arm is available for \$300–\$2,000, below any mobile or humanoid platform.
- (b) c_{env} : the tabletop is bounded, resettable, and non-safety-critical, requiring no lease, safety certification, or human subjects.
- (c) c_{int} : open-source orchestration stacks and vision-language-action models make the loop closable by a single developer in weeks.

No other entry point satisfying (E1)–(E3) achieves a lower cost in all three components. $\square$ $\square$

Remark 3.9. The theorem does not claim tabletop manipulation is the most *valuable* entry point, only the cheapest. Value is a separate criterion (Definition 3.6).

3.3 Domain Ranking

Theorem 3.10 (Domain Value Ranking). *Among entry points of equal loop cost, the ranking by domain value (Definition 3.6) is:*

- (1) *pharmaceutical compounding and lab automation,*
- (2) *electronics assembly and repair,*
- (3) *food preparation,*
- (4) *micro-fulfillment and e-commerce picking,*
- (5) *medical and dental laboratories.*

Proof. By Definition 3.6, DV increases in V , P , and T , and decreases in c_{loop} . For entry points of equal loop cost, the ranking is determined by $V \cdot P \cdot T$.

Pharmaceutical compounding and lab automation maximize each factor: V is high (expensive human labor replaced), P is maximal (trial logs are not derivable from public corpora), and T is high (fluid dynamics, material handling, and contamination avoidance transfer to adjacent domains).

Electronics assembly is close behind: V is higher (largest market), P is high, and T is high. It ranks second because the loop is somewhat more complex (more steps per cycle).

Food preparation has the highest V (largest market), but lower P and T because food is more variable and less precisely measurable.

Micro-fulfillment has high V and high T , but a larger environment (shelves, bins), raising c_{loop} above the tabletop.

Medical and dental labs have high P but smaller V and T than the others. $\square$ $\square$

3.4 Optimal Entry Strategy

Prediction 3.11 (Optimal Entry Strategy). A resource-constrained entrant should:

- (1) enter via the cheapest loop (tabletop manipulation),
- (2) target the highest-value domain within reach (pharmaceutical compounding or lab automation),
- (3) accumulate proprietary grounded data,
- (4) expand along the direction of maximum T (Definition 3.5),
- (5) extend to larger environments only after the loop is proven.

Step (4) is what converts a domain-specific integrator into a platform (Theorem 3.10, Theorem 4.7). Step (5) is the discipline that follows from staggered eras (Theorem 2.6).

Corollary 3.12 (The Arm Is Not the Product). *The tabletop arm is valuable not as a product but as an instrument for generating grounded data. By Definition 3.6, value accrues to $\mathcal{D}_A(\mathcal{E})$, not to $\mathcal{C}_{hw}$.*

Remark 3.13 (Relation to the Integrator Moat). The entry point strategy is the inverse of the integrator moat (Theorem 2.4). The moat says *where value accrues*; the entry point says *where to start*. Both follow from the same principle: grounded data is the proprietary asset, and the cheapest way to acquire it is to close the loop in a bounded environment.

4 Transfer and Generalization

The domain ranking of Section 3 selects where to close the first loop. It does not answer whether the loop can be extended beyond that domain. That question is governed by the transfer coefficient (Definition 3.5); its consequences determine the industrial shape of the grounded-agent economy.

Proposition 4.1 (Three Transfer Regimes). *Let $\mathcal{L}$ be an entry point in environment $\mathcal{E}$, and let $\mathcal{E}'$ be an adjacent environment. Then exactly one of the following holds:*

- (i) **Positive transfer.** $T(\mathcal{L}, \mathcal{E}') > 1$: grounding data from $\mathcal{E}$ accelerates grounding in $\mathcal{E}'$.
- (ii) **Null transfer.** $T(\mathcal{L}, \mathcal{E}') = 1$: grounding in $\mathcal{E}'$ must be acquired independently.
- (iii) **Negative transfer.** $T(\mathcal{L}, \mathcal{E}') < 1$: grounding data from $\mathcal{E}$ actively harms grounding in $\mathcal{E}'$.

Proof. Immediate from Definition 3.5. $\square$

Proposition 4.2 (Transfer Determines Industrial Shape). *The industrial landscape of the grounded-agent economy is a function of the transfer coefficient:*

- (1) *If $T > 1$ across many environment pairs, one loop generalizes broadly, yielding a small number of general winners.*
- (2) *If $T = 1$ across most environment pairs, every domain must be grounded separately, yielding a federation of domain-specific integrators.*
- (3) *If T is high within clusters and low between them, the landscape consists of regional winners, each dominating a modality cluster.*

Justification. By the staggered-eras result (Theorem 2.6), the robotics era is a sequence of domain-specific eras. By Theorem 2.4, each domain has an integrator moat. Whether a single integrator can occupy multiple domains depends on whether grounding data transfers from one to another. If $T(\mathcal{L}, \mathcal{E}') > 1$ for all $\mathcal{E}'$ in a set $\mathcal{S}$, then a single integrator can ground each $\mathcal{E}'$ in $\mathcal{S}$ from the loop closed in $\mathcal{E}$, and the moats in $\mathcal{S}$ are unified. If $T = 1$, the moats remain separate. If T is high within clusters, the moats unify within clusters and remain separate between them. $\square$

Definition 4.3 (Determinants of Transfer). The transfer coefficient $T(\mathcal{L}, \mathcal{E}')$ is determined by four structural factors:

- (a) **Shared physics.** Overlap in rigid-body dynamics, gravity, friction, and contact.
- (b) **Shared sensory profile.** Overlap in visual, tactile, and proprioceptive signatures.
- (c) **Shared causal structure.** Overlap in the causal relations the agent must learn (pushing $\rightarrow$ sliding, grasping $\rightarrow$ lifting).
- (d) **Shared object ontology.** Overlap in the kinds of entities the agent must represent.

The higher the overlap in each factor, the higher T .

Proposition 4.4 (Transfer Is Structural, Not Architectural). *$T(\mathcal{L}, \mathcal{E}')$ depends on the structural relation between $\mathcal{E}$ and $\mathcal{E}'$, not on the model architecture alone.*

Justification. By Definition 4.3, T is determined by overlap in physics, sensory profile, causal structure, and object ontology. These are properties of the environment pair, not of the agent. An agent with an ideal architecture but operating in environments with low overlap will have low T . An agent with a modest architecture but operating in environments with high overlap will have high T . $\square$

Corollary 4.5 (Transfer Learning Has a Structural Explanation). *Transfer learning succeeds when the four factors of Definition 4.3 overlap and fails when they do not. The success or failure of transfer is not a matter of architecture alone but of structural alignment between source and target environments.*

Prediction 4.6 (The Generalization Problem Is the Transfer Problem). Raising T is the generalization problem. All techniques that improve cross-domain transfer—shared representations, common-sense physics, compositional object models, cross-domain pretraining, embodied foundation models—are T -raisers. The first firm to raise $T > 1$ across a cluster of high-value domains will be the first to convert a narrow entry point into a platform.

Theorem 4.7 (Platform Winners Come from Transfer, Not Scale). *The platform winner in the grounded-agent economy will not be the firm with the largest model or the most capital. It will be the firm whose architecture transfers best across the environments where T is high.*

Justification. By Proposition 4.2, a general winner requires $T > 1$ across many environment pairs. By Theorem 2.5, capital cannot buy grounding. By the scaling result of Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*, model scale increases R and D but not G , and therefore does not by itself raise T , which is structural (Proposition 4.4). Hence the winner is characterized by transfer, not by scale or capital. $\square$

Remark 4.8 (Entry Strategy and Transfer). The entry strategy of Prediction 3.11 is completed by the transfer coefficient. The entrant should:

- (1) close the loop in the cheapest environment (Theorem 3.8),
- (2) select the target domain by domain value (Theorem 3.10),
- (3) expand along the direction of maximum T from the entry environment.

The third step is what converts a domain-specific integrator into a platform. It is also what determines whether the expansion path is feasible at all.

Open Problem 4.9. Is there a measurable proxy for T that does not require closing the loop in the target environment first? A proxy would allow empirical estimation of expansion paths before deployment.

5 The Structural Inversion

The framework’s results on grounding, the integrator moat, entry points, and transfer jointly entail a claim about the shape of the coming industry that is easy to miss when attention is fixed on the present. We state it here and draw out its consequences.

Definition 5.1 (Value Layer). Let an industry be represented as a stack of layers $\Lambda_1, \dots, \Lambda_m$, where each Λ_i corresponds to a distinct class of production. The *value layer* of the industry at time t is the layer $\Lambda^*(t)$ at which the majority of economic value accrues.

Definition 5.2 (Structural Inversion). A *structural inversion* occurs when $\Lambda^*(t)$ shifts from one layer to another: $\Lambda^*(t_1) \neq \Lambda^*(t_2)$ for $t_1 < t_2$.

Proposition 5.3 (Current Value Layer). *In the AI industry at present, value accrues primarily to the model layer (the generative components γ and σ).*

Justification. The largest market capitalizations, the largest capital expenditures, and the greatest concentration of research talent in the AI industry are currently directed at the model layer: foundation models, inference compute, and the training infrastructure that supports them. By Definition 5.1, this is the value layer. $\square$

Proposition 5.4 (Predicted Value Layer). *In the grounded-agent economy, value accrues primarily to the loop layer: closed epistemic loops and the grounded data they generate.*

Justification. By the generative-substitution invariance result of Application Paper I, substitution of generative components leaves G invariant, and cannot move an agent across the $5^* \rightarrow 5^+$ threshold. Grounding is environment-specific and cannot be purchased (Theorem 2.5). Hence the proprietary asset of an agent is the closed loop, which is unavailable to non-integrators (Theorem 2.4). Value therefore accrues to the loop layer. $\square$

Theorem 5.5 (Structural Inversion in the AI Industry). *Under the assumptions of the framework, the AI industry will undergo a structural inversion: the value layer will shift from the model layer (Proposition 5.3) to the loop layer (Proposition 5.4).*

Proof. Immediate from Definitions 5.1 and 5.2 and Propositions 5.3 and 5.4. $\square$

Proposition 5.6 (Invisibility of the Inversion). *The inversion is not directly visible from the present, because observers attend to the currently dominant layer.*

Justification. By Proposition 5.3, the current value layer is the model layer. Attention, capital, and discourse concentrate at the value layer. An observer attending to the model layer sees the loop layer as a supplier or a customer, not as a successor. The inversion is therefore visible only as an absence in the current map of the industry, not as a presence. $\square$

5.1 Three Strange Consequences

Prediction 5.7 (The Frontier Looks Small). The most valuable robotics company of 2040 will be narrow rather than general, boring rather than glamorous, and grounded rather than scaled.

Prediction 5.8 (Incumbent Confusion). Incumbents with the largest models, the most capital, and the widest distribution will be structurally positioned to supply the coming economy, not to win it. Capital buys components; it does not buy grounding.

Prediction 5.9 (Late and Quiet Generality). By Theorem 2.6, general-purpose robotics arrives late and diffuses quietly. There will be no single launch date for the era. There will be a diffusion curve.

5.2 The IBM Pattern

Remark 5.10 (Predictable Position, Unpredictable Player). In 1975, the structure of the coming personal computing industry was predictable: the industry would require a compute substrate, an operating system layer, and integrated consumer products. The *positions* were visible to anyone reasoning structurally. The *players* were not. IBM had more capital, more engineers, more distribution, and more brand than Microsoft and Apple combined. It lost the PC era anyway, because the winning position was not the one it occupied.

Proposition 5.11 (Same Pattern, Different Domain). *By Theorem 5.5, the same structural pattern will recur in robotics: the integrator positions will be filled, but the firms that fill them are not predictable from the current industrial landscape.*

Justification. By Theorem 2.4, the integrator position is defined by closure of an epistemic loop in an environment. The position is domain-specific and cannot be acquired by component purchase or by capital. It follows that the occupant of any given integrator position depends on which agent first closes the loop in that environment—a matter of execution, timing, and contingency that the axioms do not determine. $\square$

Corollary 5.12 (The Winners Are Unknown). *The most valuable companies of the next twenty-five years are, with high probability, not yet founded or not yet recognized.*

Proof. By Proposition 5.11, the integrator positions will be filled, but the firms that fill them are not predictable from the current landscape. By Proposition 5.6, the inversion is not directly visible from the present. Hence the future occupants of the value layer are not presently identifiable with confidence. $\square$

Remark 5.13 (This Is Not a Gap in the Framework). The framework’s inability to name the future winners is not a defect. It is the framework working as intended. A structural theory predicts the shape of a landscape, not the identity of its occupants. The PC-era framework of 1975 correctly predicted the need for a substrate, an OS, and an integrator. It did not name Microsoft, Apple, or Intel. The same restraint applies here.

5.3 The Wildest Part

Remark 5.14 (Counterintuitive and Structurally Forced). The future is wild not because it is futuristic, but because it is counterintuitive and structurally forced at the same time. The present is looking at the wrong layer. The structure of the coming industry is visible to anyone reasoning from the axioms. The occupants of its key positions are not. That is the whole story.

Prediction 5.15 (The Inversion Will Feel Sudden). Although the inversion is structurally continuous (Theorem 5.5), it will feel sudden to observers who attended to the model layer. By Proposition 5.6, the shift becomes visible only retrospectively.

5.4 Component-Layer and Value-Layer Dominance

Corollary 5.16 (Component-Layer Dominance Is Not Value-Layer Dominance). *By Theorem 5.5, the value layer of the grounded-agent economy is the loop layer. The accidental-fit firm of Application Paper I dominates the component layer. It does not occupy the value layer. Component-layer dominance and value-layer dominance are structurally distinct positions.*

Remark 5.17 (The Future Nvidia Is Not the Future Winner). The firm described by the future-Nvidia prediction of Application Paper I will be essential, profitable, and structurally positioned to supply the grounded-agent economy. It will not be the firm that occupies the value layer. The value layer is the loop (Theorem 2.4). The component layer is the supply. Confusing the two is the mistake that makes observers think the next Nvidia will be the next Microsoft. It will not. It will be the next TSMC.

6 Strategy and the Value Layer

The results of Section 5 have a direct consequence for strategy. If the value layer shifts structurally (Theorem 5.5), then strategic activity aimed at determining where value accrues is misaligned with the structure. We formalize this and distinguish futile from non-futile strategy.

Definition 6.1 (Layer-Directed Strategy). A *layer-directed strategy* for a firm Φ is a strategy whose principal objective is to make a given layer Λ_i the value layer $\Lambda^*(t)$ at some future time t .

Definition 6.2 (Loop-Directed Strategy). A *loop-directed strategy* for a firm Φ is a strategy whose principal objective is to close a loop in some environment $\mathcal{E}$.

Proposition 6.3 (Layer-Directed Strategy Does Not Move the Value Layer). *No layer-directed strategy changes the value layer.*

Proof. By Theorem 5.5, the value layer shifts as a structural consequence of grounding: from the model layer to the loop layer. This shift is determined by Definitions 5.1 and 5.2, together with Propositions 5.3 and 5.4. None of these depends on the strategies of individual firms. Hence no firm's layer-directed strategy can change the value layer. $\square$

Proposition 6.4 (Loop-Directed Strategy Is Structurally Aligned). *Loop-directed strategy is the only strategy aligned with the structural shift of Theorem 5.5.*

Proof. By Proposition 5.4, the value layer in the grounded-agent economy is the loop layer. By Theorem 2.4, the asset that accrues to loop-closers is grounded data, which is non-derivable and non-purchasable (Theorem 2.5). A firm that pursues a loop-directed strategy is therefore building the asset that defines the value layer. No other strategy targets this asset. $\square$

Corollary 6.5 (Three Futile Activities). *The following strategic activities are futile under the framework:*

- (i) **Dictating the value layer.** *Attempting to make the model layer permanently dominant (Proposition 6.3).*
- (ii) **Naming the occupants.** *Attempting to determine, by acquisition, partnership, or ecosystem lock-in, which firm occupies the loop layer (Corollary 5.12).*
- (iii) **Buying grounding.** *Attempting to acquire, by purchase, the asset that defines the value layer (Theorem 2.5).*

Corollary 6.6 (Three Non-Futile Activities). *The following strategic activities are not futile under the framework:*

- (i) **Occupying the loop layer.** *Pursuing a loop-directed strategy, which is structurally aligned (Proposition 6.4).*

(ii) **Building loop-closure tools.** Standardizing loop closure across a domain.

(iii) **Funding the people who will build the next layer.** Treating a firm's alumni as its real export.

Theorem 6.7 (The Sharpest Formulation). *Current model-layer firms are attempting to direct the future by controlling the layer that the structure is moving away from. This is not directing the future. It is spending the present to hold a position from which the value layer is departing.*

Proof. By Proposition 5.3, model-layer firms occupy the current value layer. By Theorem 5.5, the value layer is shifting to the loop layer. By Proposition 6.3, no layer-directed strategy changes this. Hence any model-layer firm attempting to make the model layer permanent is spending resources to hold a position that the structure is leaving. $\square$

Prediction 6.8 (Incumbent Strategy Will Be Misaligned). Incumbent AI firms will invest, in the near term, in layer-directed strategies: scaling models, integrating model layers, acquiring model-layer assets, and locking in ecosystem positions at the model layer. These investments will not move the value layer. They will be demoted along with the layer they are directed at.

Prediction 6.9 (Aligned Strategy Will Come from Outside). The firms that will occupy the loop layer will not be the current model-layer incumbents. They will come from outside the current landscape—from components, from adjacencies, from new entrants—for the same structural reason that the occupants of the PC value layer came from outside the semiconductor incumbency.

Remark 6.10 (Not Doom, Demotion). The framework does not predict that current AI leaders will fail. It predicts they will be demoted from the value layer to the component layer. Intel and TSMC are demoted relative to Microsoft and Apple, and they remain among the most consequential firms in the world. The terrifying part is not survival. It is the loss of the assumption that the current layer is where the future lives.

Remark 6.11 (The Assumption Under Attack). The assumption that the current layer is where the future lives is the assumption most current AI strategy acts on. The framework says the assumption is structurally false. It is not false as a matter of opinion or of forecasting. It is false because the value layer is determined by grounding, and grounding does not accrue to the model layer. That is the sharpest claim the framework makes about the current moment.

7 Open Problems

Open Problem 7.1. Under what conditions does grounding in $\mathcal{E}_1$ partially transfer to $\mathcal{E}_2$? What is the transfer function?

Open Problem 7.2. Is there a closed form for the transfer coefficient $T(\mathcal{L}, \mathcal{E}')$? What modality structure is required for $T > 0$ between two environments? This would allow quantitative prediction of domain expansion paths.

Open Problem 7.3. What is the rate of the structural inversion (Theorem 5.5)? Can the arrival time of the loop layer be estimated?

Open Problem 7.4. The market projections of the appendix (Section I) provide empirical inputs to $V(\mathcal{L})$. Does the framework impose measurable constraints on their mutual consistency—for instance, do the robot-count and energy figures cohere with the groundable labor base?

Remark 7.5. The predictions of this paper are consequences of the axioms of Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*, not forecasts about engineering timelines or market outcomes. They state what follows *if* an agent or firm has the stated properties, not when or whether such agents or firms will exist.

8 Conclusion

We have applied the inference-first framework to entry points, domain value, transfer, and the structural inversion of the AI industry. The central results are:

1. The domain value $DV = V \cdot P \cdot T / c_{\text{loop}}$ scores an entry point by commercial value, data propriety, transfer, and loop cost (Definition 3.6).
2. The cheapest general entry point is a tabletop manipulation loop with a low-cost robotic arm (Theorem 3.8).
3. Among equal-cost entry points, pharmaceutical compounding and lab automation rank highest by domain value, followed by electronics assembly, food preparation, micro-fulfillment, and medical laboratories (Theorem 3.10).
4. The transfer coefficient has three regimes: positive, null, and negative (Proposition 4.1).
5. The industrial shape of the grounded-agent economy is a function of T (Proposition 4.2).
6. T is determined by four structural factors—shared physics, sensory profile, causal structure, and object ontology—not by architecture alone (Definition 4.3, Proposition 4.4).
7. Raising T is the generalization problem, and the first firm to raise $T > 1$ across a cluster of high-value domains will be the first to convert a narrow entry point into a platform (Prediction 4.6, Theorem 4.7).
8. The entry strategy is the inverse of the integrator moat: value accrues to whoever closes the loop first, and the cheapest way to close it is in a bounded environment (Prediction 3.11).
9. The arm is not the product; the grounded data is (Corollary 3.12).
10. The AI industry will undergo a structural inversion: the value layer shifts from the model layer to the loop layer (Theorem 5.5).
11. The inversion is not directly visible from the present, because observers attend to the currently dominant layer (Proposition 5.6).
12. The most valuable companies of the next twenty-five years are, with high probability, not yet founded or not yet recognized (Corollary 5.12). This is not a gap in the framework; it is the framework working as intended (Remark 5.13).
13. No layer-directed strategy changes the value layer (Proposition 6.3); loop-directed strategy is the only aligned strategy (Proposition 6.4).
14. Three activities are futile; three are not (Corollary 6.5, Corollary 6.6).
15. Current model-layer firms are spending the present to hold a position from which the value layer is departing (Theorem 6.7). This is not doom but demotion (Remark 6.10).

Applied to the present moment, the framework yields a sharp prediction: model, orchestration, and hardware layers commoditize; the closed loop does not. The proprietary asset is the grounded data generated by lived interaction in a specific environment. The cheapest path to that asset is a narrow loop in a bounded environment, expanded along the direction of maximum transfer. The largest markets are destinations, not on-ramps; the entry point is the strategy, and the transfer operator decides how far the strategy travels.

Part I

Appendix: Market Projections and Reconciliation

The following appendix is empirical rather than formal. It reports long-range market projections to instantiate the components of commercial value (Definition 3.4) and to illustrate the domain ranking (Theorem 3.10). The projections are as reported by the cited analysts, on a roughly 2045–2055 horizon. The framework itself does not generate these numbers; it prescribes how they should be used: to filter domains by loop cost and data propriety after ranking by size.

9 Physical AI and Robotics Verticals

Ranked by projected market size in the 2045–2055 window. Figures are the most recent long-range projections available.

Rank	Domain	Projected (2050)	Size	Key Driver
1	Robotics (all physical systems)	\$25T (hardware alone)		Morgan Stanley GROM: 1.4B units sold annually, 6.5B robots in operation; 90% for industrial/commercial use
2	Humanoid Robots	\$5T–\$9T		1B humanoids by 2050; RBC: \$9T TAM including \$3T software ecosystem; China 61% of market
3	Autonomous Vehicles & Robotaxis	7.22B vehicles in operation		98% of global new car sales autonomous by 2050; 100M AVs on U.S. roads alone
4	Logistics & Warehousing	\$5T (segment of \$25T)		90% of humanoids deployed for industrial applications; 75% warehouse automation by 2050; 25–30% productivity gains already proven
5	Agriculture (AgriTech)	\$250B (AI agriculture market)		10B people to feed by 2050; autonomous robot fleets in fields; precision spraying reduces pesticide use 10x
6	Aerospace & Defense	\$162B (CCA segment alone)		AI-enabled drones flying alongside crewed fighters; autonomous air and space operations routine by 2050
7	Healthcare & Life Sciences	\$1.4T–\$1.7T (humanoid TAM)	(humanoid TAM)	1B AI-enabled humanoid robots by 2050 at \$5T total market; 300M humanoids in workplace by 2050
8	Energy & Utilities	£8.4B (robotics segment)		Autonomous inspection of grids, solar farms, plants; 500M active robots by 2050 with energy impact rivaling AI data centers
9	Construction & Infrastructure	\$11.14B by 2040; accelerating beyond		13,000 buildings needed daily by 2050; sites 60% machines, 40% human by mid-century

Sub-Domain Highlights

- **Humanoids:** RBC projects \$9T TAM by 2050, including \$2.9T in households (33% of total). China expected to capture 61% of the market. Software app ecosystems could add \$3T in value.
- **Agriculture:** The AI agriculture market alone could reach \$250B by 2050. Autonomous robot fleets are expected in fields by 2050. Precision spraying already reduces pesticide use by a factor of 10.
- **Logistics:** Amazon's fleet of over 1M robots demonstrates the closed loop at scale. Morgan Stanley projects \$5T in this segment alone within the broader \$25T robotics market.
- **Energy:** Robotics in energy is projected at £8.4B by 2050, with oil and gas peaking at £5B in 2030 before declining to £3.3B by 2050.

10 AI Agents and Software-Dominated Services

Ranked by addressable market. These figures are TAM, not the eventual AI-segment capture, and are therefore larger than the physical-robotics projections above.

Rank	Domain	Addressable Market	Automation Focus
1	Legal Services	\$400B+	By 2050, AI has replaced much of what lawyers called legal work: drafting, reviewing, predicting, pricing, organizing, authenticating, reporting. 50% decline in legal positions projected by 2050
2	Insurance & Financial Services	\$140–200B (brokerage alone)	80% of routine fraud detection automated by 2035; fully automated claims and AI-driven underwriting by 2050–2060
3	IT Managed Services	\$100B+	Shift from cybersecurity-led to AI-driven managed services. 500 active agents can replace 50 human technicians. Managed services market projected at \$595B with AI as primary growth vector
4	Healthcare Revenue Cycle	\$900B in savings (not revenue pool)	AI could save hospitals \$900B by 2050. RCM revenue pool could shrink 50% by 2040 as AI automates prior authorization, documentation, billing
5	Customer Service & Support	Mature segment; 85% query automation	Tier-1 support, triage, resolution. 25–30% cost reduction already demonstrated in production deployments
6	Software Engineering	No consolidated TAM; already profitable	Development, debugging, code review. AI agents already generating revenue at scale in coding assistance

11 Why the Two Rankings Are Separate

The figures in the two tables are not directly comparable:

- **Physical-robotics figures** report projected market size for physical systems. Morgan Stanley’s \$25T figure is hardware sales alone; with software, maintenance, and supply chain, the addressable market may be several times larger.
- **Agent figures** report addressable market. This is the total spend in the sector, not the portion that will be captured by AI agents. The capturable fraction is typically 5–15%.
- **Maturity differs.** Software agents are already generating revenue at scale. Physical AI is earlier in its adoption curve, with mass deployment of humanoids expected toward the end of the 2030s.

A unified ranking would be misleading. On a 10% capture assumption, legal services yields a \$40B AI opportunity, insurance \$14–20B, IT services \$10B, and healthcare revenue cycle would be a cost-savings play rather than a revenue opportunity. The physical-robotics figures, by contrast, are direct revenue projections.

12 The 20–30 Year Trajectory

Over the next 20–30 years, the physical-robotics domains will dominate by absolute revenue. Morgan Stanley’s \$25T hardware projection for 2050 dwarfs any software-agent TAM. The trajectory is driven by five factors:

1. **Cost curves.** Average unit price of a robot is projected to fall from \$200,000 (2024) to \$50,000 by 2050 in developed markets, and as low as \$15,000 in lower-income countries.
2. **Labor replacement.** By 2050, humanoids could replace up to 40% of labor-intensive roles across agriculture, manufacturing, and cleaning.
3. **China’s dominance.** China is expected to account for roughly 61% of the humanoid market and 26% of global robot sales by 2050, driven by manufacturing scale and policy support.
4. **Software flywheel.** RBC projects an additional \$3T in software value through app store ecosystems where users download capabilities ranging from caregiving to meal preparation.
5. **Energy constraints.** 500M active robots by 2050 will consume energy at a scale rivaling AI data centers, creating both a constraint and an opportunity for energy-domain integrators.

13 Reconciling the Ranking with the Framework

By the framework’s own criteria, the most lucrative domains are not determined by market size alone. They are determined by the combination of loop cost, data propriety, commercial value, and transfer (Definition 3.6). A large market with a high loop cost and low data propriety is less attractive than a moderate market with a cheap loop and high data propriety. The rankings above should be read as a starting point, not a conclusion.

The reconciliation with the entry strategy (Prediction 3.11) is direct:

- **Do not chase rank 1.** Robotics at \$25T is the aggregate market, not an entry point. No startup enters at rank 1.

- **Enter at the intersection of the cheapest loop and the highest domain value.** That intersection is pharma compounding and lab automation—not because they top the market-size list, but because they combine low loop cost with high data propriety and high transfer (Theorem 3.10, Theorem 3.8).
- **Expand toward the larger markets.** Once the loop is proven, the transfer coefficient determines which ranked domains are reachable. Pharma $\rightarrow$ electronics $\rightarrow$ food is the natural expansion path.
- **The largest markets are reached last, not first.** The \$25T robotics market is the destination. The \$2,000 tabletop arm is the on-ramp. Between them lies a decade of accumulated grounded data.

Remark 13.1 (Map and Compass). The ranking is a map of where value will be. The entry strategy is how to get there. Neither is useful without the other; the map without the strategy is unactionable, and the strategy without the map is directionless.

Rank by market size. Filter by loop cost. Choose by data propriety. Win by transfer.

A Summary of Definitions (Application-Specific and Imported)

Symbol	Name	Reference
$\mathcal{L}$	Entry point loop	Definition 3.1
$c_{\text{loop}}(\mathcal{L})$	Loop cost	Definition 3.2
$P(\mathcal{L})$	Data propriety	Definition 3.3
$V(\mathcal{L})$	Commercial value	Definition 3.4
$T(\mathcal{L}, \mathcal{E}')$	Transfer coefficient	Definition 3.5
$DV(\mathcal{L})$	Domain value	Definition 3.6
Transfer determinants	Physics, senses, causality, ontology	Definition 4.3
$\Lambda^*(t)$	Value layer	Definition 5.1
Structural inversion	Value layer shift	Definition 5.2
Layer-directed strategy	Objective is to make a layer the value layer	Definition 6.1
Loop-directed strategy	Objective is to close a loop	Definition 6.2

B Summary of Theorems (Application-Specific and Imported)

Theorem	Statement
3.8	Cheapest general entry point
3.10	Domain value ranking
4.1	Three transfer regimes
4.2	Transfer determines industrial shape
4.4	Transfer is structural, not architectural
4.5	Transfer learning has a structural explanation
4.7	Platform winners come from transfer, not scale
3.12	The arm is not the product
5.3	Current value layer is the model layer
5.4	Predicted value layer is the loop layer
5.5	Structural inversion in the AI industry
5.6	Inversion invisible from the present
5.11	Same IBM pattern, different domain
5.12	Winners are unknown
5.16	Component-layer $\neq$ value-layer dominance
6.3	Layer-directed strategy does not move the value layer
6.4	Loop-directed strategy is structurally aligned
6.5	Three futile activities
6.6	Three non-futile activities
6.7	The sharpest formulation

C A Note on Method

This paper proceeds by applying the formal framework of Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*. Its formal results are conditional and framework-relative. Its predictions are consequences of the axioms, not forecasts about engineering timelines or market outcomes. The market projections in the appendix are empirical inputs, reported as published by the cited analysts; they instantiate, but are not generated by, the framework.